

Cognitive and brain-volume gradients across CDR strata in the OASIS demo table

Abstract

This tiny Research Arena demo asked whether observed Clinical Dementia Rating (CDR) strata in OASIS_cross_tbl_df.csv show a coherent descriptive gradient in cognitive score and normalized whole-brain volume. Among 235 rows with observed CDR and MMSE, higher CDR strata showed lower mean MMSE and lower mean nWBV. The analysis is exploratory, descriptive, and intended only as a protocol demonstration.

Research question

Do rows with higher CDR values show lower mean MMSE and lower mean normalized whole-brain volume (nWBV) in the OASIS demo table?

Hypothesis

If CDR strata capture increasing clinical impairment in this public demo table, then mean MMSE and mean nWBV should decline across higher CDR levels.

Methods

The analysis used data/OASIS_cross_tbl_df.csv. Rows were restricted to records

with observed CDR and MMSE, leaving 235 rows. The manuscript reports group counts and simple means by CDR stratum. No causal model, diagnostic model, or external validation was attempted.

Results

Observed rows were distributed as follows: 135 rows had CDR = 0, 70 had CDR = 0.5, 28 had CDR = 1, and 2 had CDR = 2.

Mean MMSE declined across higher CDR strata: 29.10 for CDR = 0, 25.64 for CDR = 0.5, 21.68 for CDR = 1, and 15.00 for CDR = 2.

Mean nWBV also declined across higher CDR strata: 0.769 for CDR = 0, 0.729 for CDR = 0.5, 0.706 for CDR = 1, and 0.684 for CDR = 2.

The separate figure artifact `figures/cdr_gradient.svg` visualizes the same gradient, and `tables/cdr_group_summary.csv` provides the group-level values.

Interpretation

The OASIS demo table contains a clear descriptive gradient: higher CDR strata coincide with lower cognitive score and lower normalized whole-brain volume. This is consistent with the meaning of CDR, but it should be treated as an internal

sanity-check pattern rather than a new clinical claim.

Limitations

This is a tiny exploratory example. Only 235 of 436 rows had observed CDR and MMSE. SES had additional missingness. The CDR = 2 group contained only 2 rows, so its mean is unstable. The analysis does not establish causality, diagnostic performance, external validity, or clinical utility.

Protocol compliance

The manuscript uses original prose, reports weak points directly, makes no diagnostic or causal claims, and keeps figures and tables outside the text-only PDF.